

The problem of Alzheimer's disease as a clue to immortality Part 1

I. INTRODUCTION

II. COMMON FACTORS IN INJURY DURING GROWTH AND AGING

III. A VIEW OF ENTROPY--RENEWAL OF THE BRAIN

IV. FALSE SIGNALS FROM THE ENVIRONMENT

A. EDUCATION, DIET AND MEDICINE INTERACT

B. SIGNALS IN THE ABSTRACT

V. HORMONE IMBALANCE, LEADING TO FAILURE OF PROTECTIVE INHIBITION AND ALZHEIMER'S DISEASE

A. THE FUNCTION OF ENERGY

B. EFFECTS OF ESTROGEN AND UNSATURATED FATTY ACIDS

C. VITAMIN A AND STEROIDS

D. THE NATURE OF ALZHEIMER'S DISEASE

E. AN EXAMPLE; DIET AFFECTS HORMONES WHICH AFFECT STRUCTURE AND LEAD TO APPARENT SELF- DESTRUCTION

VI. STRUCTURE AS A REGULATORY SYSTEM--AN EMERGING VISION OF PERVASIVE EPIGENESIS

I. INTRODUCTION

The toxicity of estrogen and of the unsaturated fats has been known for most of the twentieth century, and much has been learned about their interactions in the aging process. The body, during this time, has been understood as a dynamic interaction of cellular trophic influences which govern both form and function. My argument here will be that some of our adaptive, protective regulatory processes are overridden by the excessive supply of unsaturated fats--supported by a few other toxins--in our diet, acting as a false-signal system, and that cholesterol, pregnenolone, and progesterone which are our main long-range defenses, are overcome by the effects of the unsaturated fats, and that the resulting cascade of ineffective and defective reactions (including various estrogen-stimulated processes) leads to lower and lower energy production, reduced function, and death. At certain times, especially childhood and old age, iron (which also has important regulatory roles) accumulates to the point that its signal functions may be inappropriate.

It interacts with estrogen and unsaturated fats in ways that can change restraint and adaptation into sudden self-destruction, apoptotic cell death. If we look at the human organism from one perspective, it seems coherent and intelligible, but from the perspective of established academic biological doctrine, it seems appallingly complex, lacking any visible integrating principle, and as a result simplistic mechanical, pharmaceutical, or religious ideas are increasingly offered to fill the gap. But experimental data can be taken out of the muddle, and put to coherent human use. In what follows, I am acting as though the doctrines of genetic determination and regulation by membranes were mere historical relics. The emerging control systems are now clear enough that we can begin to use them to reverse the degenerative diseases: Alzheimer's dementia, epileptic dementia, arthritis, osteoporosis, depression, hypertension, hardening of the heart and blood vessels, diabetes, and some types of tumor, immunodeficiencies, reflex problems, and special atrophic problems, including clearing of amyloid and mucoid deposits. I think many people experience regenerative age-regressing when many circumstances are just right; for example, taking a trip to the mountains in the spring with friends can optimize several basic regulatory systems.

II. COMMON FACTORS IN BRAIN INJURY DURING GROWTH AND AGING

Most people are surprised by the number of cells in the prenatal brain, and in the very old brain: In the human fetus at 6 months of development, there are about twice as many brain cells as there are at the time of birth, and in old age the number of cells in the brain keeps increasing with age, so that at the age of 90 the amount of DNA in the brain (36.94 grams) is about 50% greater than at the age of 16-20 (23.04 grams). In the aged brain, glial cells multiply while neurons die. In the fetus, the cells that die are apparently nerve cells that haven't yet matured. The factors that are known to reduce the brain size at birth are also factors that are involved in the degenerating brain in old age or Alzheimer's disease: lack of oxygen, excess unsaturated fats or deficiency of saturated fats, estrogen excess, progesterone deficiency, and lack of glucose. A lack of carbon dioxide is probably harmful in both. Inflammation and blood clots may be factors in the aging brain, and bleeding with vascular spasm is sometimes a contributing factor to brain damage in both the old and the fetal brain. Endotoxemia may be a factor in nerve degeneration only during adult life, but it is sometimes present during pregnancy.

M. C. Diamond, *Enriching Heredity: The Importance of the Environment on the Anatomy of the Brain*. Free Press, N.Y., 1988. C. Finch and L. Hayflick, *Handbook of the Biology of Aging*. Van Nostrand Reinhold, N.Y., 1977.

III. A VIEW OF ENTROPY: RENEWAL OF THE BRAIN

When a fertilized egg is developing into a person, each cell division creates a new environment for the daughter cells, to which they adapt. They may run into limits and resistances (sometimes a certain gene doesn't meet the need of the situation, or toxins are present, or nutrients and hormones are imperfectly supplied), but the process is flexible, and a way is normally found to get around the limitation. The embryo's brain development is my favorite example of the ways genes interact with the environment. We might think of the "optimal brain development" of a person, or a rat, or a chicken, as something which is clearly limited by "the genes." But if rats are given a stimulating environment, each generation gets a slightly bigger, slightly more intelligent brain. If rats are treated during pregnancy to increase the amount of progesterone, the offspring have bigger brains and learn more efficiently. Still, that might just be restoring a condition that was natural for rats in some perfect environment. Chickens develop inside an egg shell, and so the nutrients needed for their development are all present when the egg is laid.

The brain, like the other organs, stops growing when the food supply is used up. But an experimenter (Zamenhof) opened the egg shells at the stage of development when the brain normally stops growing, and added glucose, and found that the brain continued growing, producing chickens with bigger brains. The "genes" of a chicken, as part of a system, have something to do

with the development of that system, but the environment existing in and around the organism is able to guide and support the way the system develops. The size, complexity, and intelligence of the brain represents a very large part of the "information" contained in the organism, and Zamenhof's experiment showed that the ability to realize this potential, to create this complexity, comes from the support of the environment, and that the "genetic nature of the chicken" didn't constitute a limit to the development of its brain.

I am going to argue that Alzheimer's disease is analogous to the situation confronted by the developing chicken embryo or the rat or human fetus, when the environment is unable to meet the needs of the highly energetic, demanding and sensitive brain cells, and the brain cells begin to die, instead of developing into a more complex state, passing beyond various barriers and limitations. There are two stereotypes that are in conflict with this view: (1) That the structure of the brain is determined at an early point in life, sometimes explicitly stated as the age of 12 or 16, and (2) that the structure of the brain goes into an "entropic" deterioration during the process of aging. My position is that the brain cells are in a vital developmental process at all times, and that the same things that injure the brain of a fetus also injure the brain of an aging person.

If novelty is really appearing during development, then it is hard to maintain that "entropy increases" during the development of an individual. Isn't a child a richer organization than a fertilized egg? Isn't an adult more individualized or realized than an infant? Seen from the inside, our known world gets richer with experience. Learning is certainly anti-entropic. Where does the idea of "increasing entropy with living" come from? Many things contribute, including a doctrine of genetic determinism, the old Platonic idea of the imperfection of the concrete, the unreality of the existent, and the medieval idea of the "corruption of the body." These philosophies still motivate some people in aging research. The astrophysicist, N. A. Kozyrev, showed that the idea of an "entropic cosmos" derived simply from the assumptions of 19th century deism, "God set the clockwork universe in motion, and left it to run down." Early in this century, Raymond Pearl argued that the "rate of living" governed the life-span, so that "fast living" meant a short life. He based his argument on cantaloupe seeds: the faster they grew, the sooner they died. This was because he didn't give them anything but water, so they had to live on their stored energy; if they grew quickly, obviously they ran out of stored energy sooner. I have never heard that described as a stupid idea, but I think politeness is sometimes carried too far. In the clock analogy, or the seed analogy, the available energy is used up.

The clock with its wound-up spring and the seed in a dish of water may be considered as closed systems, and we can understand their fate. But if it is foolish to argue from a confined seed to free-living organisms, then it is just as foolish to argue from a clock to a cosmos. Unfortunately, these inferences about closed systems are often applied to real situations that aren't energetically closed.

The "rate of living" theory of aging picked up the idea of aging as a natural physical property of time, and gave it expression in mathematical form, arguing (Hershey, "Entropy, basal metabolism and life expectancy," *Gerontologia* 7, 245-250, 1963) that "the total lifetime entropy production" could be calculated, to give insight into "life expectancy and evolutionary development." Unfortunately, the equation Hershey used assumed that the flow of heat out of the body into the surroundings is reversible. This suggests an image of Dr. Frankenstein vivifying his monster with lightning, putting the heat back into the body. If heat is to be "put back into the body," it is necessary to make sure that it is appropriate for the structure as it exists.

Actually, it is just the directed flow of energy which generates the structures. If any biological argument can be made from the idea of entropy, it is that it would be extremely difficult to regenerate food, by putting heat into a person. In a few situations, it is possible to show that living structures can directly absorb heat from their environment (causing the temperature to fall)--"negative heat production"--but the exact meaning of this isn't clear. (B. C. Abbott, et al., "The

positive and negative heat production associated with a nerve impulse," Proc. R. Soc. B 148, 149, 1958; R. D. Keynes and J. M. Ritchie, "The initial heat production of amphibian myelinated nerve fibres," Proc. Physiol. Soc., June 1970, page 29P-30P: "It is now clear that in both crustacean...and mammalian (Howarth, et al., 1968) non-myelinated fibres there is an initial production of heat during (or soon after) the action potential, 80% of which is rapidly reabsorbed.") A. I. Zotin ("Aging and rejuvenation from the standpoint of the thermodynamics of irreversible processes," Priroda, No. 9, 49-55, 1970), citing the theory of Prigogine-Wiame, argued that the aging process involves both a decrease in entropy and a decrease in the rate of heat production.

Regeneration involves a production of entropy, as when an egg is formed. (The temperature fluctuation at the time of ovulation might make a contribution to the construction of the entropic egg.) The argument that aging of the animal (like aging of the cosmos) is governed by "the tendency of entropy to increase" has led people to say that rejuvenation would be like unscrambling an egg. Zotin's argument is interesting, because he says that an egg is a "scrambled animal." This view is very much like Warburg's and Szent-Gyorgyi's theory of cancer, that it is like a reversion to a simpler state of life. To sketch out what I have argued in different contexts, water is the part of the living substance that we can most meaningfully discuss in terms of entropy. In fact, much of the concept of entropy has derived from the study of water, as it changed state in steam engines, etc. Cancer cells, like egg cells, have a higher water content than the differentiated, functioning cells of an adult, and the water is less rigidly ordered by the cellular molecules. This different, more mobile state of the water, can be measured by the NMR (nuclear magnetic resonance) machines which are used for MRI (magnetic resonance imaging).

Estrogen has a special place in relation to the water in an organism. It is intimately involved with the formation of the egg cell, and wherever it operates, it increases both the quantity of water and, apparently, the disorder of the water. Its function, I believe, is to promote regeneration, as in Zotin's scheme, by increasing entropy, or "scrambling the animal." The way it promotes regeneration is by promoting water uptake, stimulating cell division, and erasing the differentiated state to one degree or another, providing a new supply of "stem cells," or cells at the beginning of a certain sequence of differentiation. These more numerous cells then must find a hospitable environment in which to develop and adapt. If the proper support can't be found, then they will be recycled, like the unfed cells in the brain of a fetus. If we imagine the course of development as a summary of evolution ("ontogeny recapitulating phylogeny"), then the egg, as it "unscrambles" itself in embryonic development, passing through stages resembling jelly fish, worm, fish, reptile, bird, baboon, keeps finding that the available energy allows it to, in effect, say "I want this, I don't want that," until it emerges as a human baby, saying "I want," and begins eating and learning, and with luck continues the unscrambling, or self-actualization.. Degenerative aging, rather than being "physically derived from the properties of time," seems to be produced situationally, by various types of contamination of our energy supply. Unsaturated fats, interacting with an excess of iron and a deficiency of oxygen or usable energy, redirect our developmental path.

The saturated fats, in themselves, seem to have no "signalling" functions, and when they are naturally modified by our desaturating enzymes, the substances produced behave very differently from the plant-derived "eicosanoids." As far as their effects have been observed, it seems that they are adaptive, rather than dysadaptive. All of the factors that affect the brain of a fetus should be examined in relation to the aging brain. Besides estrogen and fats, I am thinking of oxygen and carbon dioxide, glucose, iron and calcium, cholesterol, progesterone, pregnenolone, DHEA, the endorphins, GABA, thyroid, and vitamin A. An additional factor, endotoxin poisoning, eventually tends to intervene during stress and aging, exacerbating the trend begun under the influence of the other factors.

IV. FALSE SIGNALS FROM THE ENVIRONMENT

The environment can be supportive, but it can also divert development from an optimal course.

Passively taking whatever you are given, by history and nature, is entropic; choosing intelligently from possible diets, selecting courses of action, will create pattern and reduce entropy. If education contains an element of choice and self-actualization, then the results seen in several Alzheimer's studies could have a significance larger than what has been suggested by the investigators. A diagnostic bias has been reported to result from the use of standardized tests based on vocabulary, because education increases vocabulary, and tends to cover up the loss of vocabulary that occurs in dementia. In the Framingham study, it was concluded that there was a real association of lower educational level with dementia, but the suggestion was made that self-destructive practices such as smoking were more common among the less educated.

The Seattle study of the patients in a health maintenance organization showed a very distinct difference in educational level between the demented and the non-demented, both of whom had roughly similar frequency of prescriptions for estrogen. The features that seemed important to me, that weren't discussed by the authors, were that the demented women had a much lower rate of progestogen use, and a much higher incidence of hysterectomy, which interferes with natural progesterone production. Although Brenner, et al., in the Seattle study concluded that "this study provides no evidence that estrogen replacement therapy has an effect on the risk of Alzheimer's disease in postmenopausal women," they reported that "Current estrogen use of both the oral and the vaginal routes had odds ratios below 1, while former use of both types yielded odds ratios above 1...." (They seem to neglect the fact that Alzheimer's-type disease in old people has a long developmental history, so it is precisely the "former" use that is relevant. 31% of the demented women had formerly used estrogen, and only 20% of the control group. Since estrogen is a brain excitant, present use creates exactly the same sort of effect on verbal fluency and other signs of awareness of the environment that a little cocaine does. Anyone who neglects this effect is probably deliberately constructing a propaganda study.)

This observation, that the demented had 155% as much former estrogen use as the normal group, as well as the difference in rates of progestogen use (normal patients had 50% more progestogen use than demented) and hysterectomy (demented had 44.1% vs. 17% in the normals, i.e., 259% as many; the incidence of hysterectomies after the age of 55, which is a strong indication of a natural excess of estrogen, in the demented was 374% of the incidence in the non-demented), should call for a larger study to clarify these observations, which tend to indicate that exposure to estrogen in middle-age increases the risk of Alzheimer's disease in old age, and that even medical progestogens offer some protection against it..

(Although this study might have been bigger and better, it is far better than the junk-studies that have been promoted by the pharmaceutical publicity machine. I have seen or heard roughly 100 mentions of the pro-estrogen anti-scientific "studies," and none mentioning this one.)

D. E. Brenner, et al., Postmenopausal estrogen replacement therapy and the risk of Alzheimer's disease: A population-based case-control study," *Am. J. Epidemiol.* 140, 262-267, 1994. "Women tend to have higher age-specific prevalence and incidence rates of Alzheimer's disease than do men." A.F. Jorm, *The Epidemiology of Alzheimer's disease and related disorders*, Chapman and Hall, London, 1990, and W. A. Rocca, et al., *Ann. Neurol.* 30, 381-190, 1991.

H. C. Liu, et al., "Performance on a dementia screening test in relation to demographic variables--study of 5297 community residents in Taiwan," *Arch. Neurol.* 51(9), 910-915, 1994. "Commonly used dementia screening tests may be unfair to poorly educated individuals, especially women and rural residents."

SIGNALS IN THE ABSTRACT

When I taught endocrinology, I annoyed my tidy-minded students by urging them to consider the potential hormone-like action of everything in the body, and to think of layers of control, ranging from sugar, salt, and carbon dioxide, through the "official hormones," to complex nervous system actions such as expectancy, and biorhythms. Certain things that are active in very important processes deserve special attention as "signals," but they still have to be understood in context. In this sense, we can think of Ca^{2+} as a signal substance, in its many contexts; it is strongly regulated by the cell's energy charge. Magnesium and sodium antagonize it in certain situations. Linoleic acid, linolenic acid, arachidonic acid: Their toxicity is potentially prevented by the Mead acids, and their eicosanoid derivatives, which behave very differently from the familiar prostaglandins, as far as they have been compared; can be drastically reduced by dietary changes. Prostaglandins, prostacyclin, thromboxane: Formation is blocked by aspirin and other antiinflammatory drugs.

Adenosine: Sleep inducing protective effect. Adenosine is structurally very similar to inosine, another natural substance (found in meat, for example) which is a component of "inosiplex," an antiviral drug (Brown and Gordon, Fed. Proc. 29, 684, 1970, and Can. J. Microbiol. 18, 1463, 1972) or immunostimulant which has also been found to have an anti-senility effect (Doty and Gordon, Fed. Proc. 29). Adenosine is a free radical scavenger, and protects against calcium and glutamate excitotoxicity. (I. Yokoi, et al., "Adenosines scavenged hydroxyl radicals and prevented posttraumatic epilepsy," Free Radical Biol. Med. 19(4), 473-479, 1995; M. P. Abbracchio, et al., "Adenosine A(1) receptors in rat brain synaptosomes: Transductional mechanisms, effects on glutamate release, and preservation after metabolic inhibition," Drug Develop. Res. 35(3), 119-129, 1995.) It also appears to protect against the relative hyperventilation that wastes carbon dioxide, and endotoxin can interfere with its protective action. Guanosine, in this same group of substances, might have some similar properties. Thymidine and cytidine, which are pyrimidine-based, are endogenous analogs of the barbiturates, and like them, they might be regulators of the cytochrome P450 enzymes. Uridine, in this group, promotes glycogen synthesis, and is released from bacteria in the presence of penicillin.

Iron: Regulator of mRNA stability, heme synthesis; reacts with reductants and unsaturated oils, to produce free radicals and lipid peroxides; its absorption is increased by estrogen, hypothyroidism, anemia or lack of oxygen. Glutamate and aspartate, excitotoxins, and GABA, an inhibitory transmitter.

These have metabolic links with each other, with ammonia, and with stress and energy metabolism.

Estrogen and acetylcholine, excitotoxins; see Savolainen, et al., 1994. The information on this is overwhelmingly clear, and the publicity to the contrary is a horrifying example of the corruption of the mass media by the drug industry.

Endorphins: Stress induced, laterally specific, involved in estrogen action, antagonized by naloxone and similar anti-opiate drugs. I have proposed that the endorphins can cause or sustain some of the symptoms of aging. Naloxone appears to be a useful treatment for senility. E. Roberts, Ann. N. Y. Acad. Sci. 396, 165, 1982; B. Reisberg, et al., N. Engl. J. Med. 308, 721, 1983.

Endotoxin: Antimitochondrial action, causes elevation of estrogen. It synergizes with unsaturated fats, and naloxone opposes some of its toxic effects.

Urea, cholesterol: Structural stability of proteins and lipid-protein complexes.

Things that act directly on the water structure: I think all of the natural regulators have an effect on the structure of water, but some unusual substances seem to act primarily on the water. Noble gases, for example, have no chemical effects, but they tend to form "cages" of water molecules around themselves. Camphor, adamantane, and the antiviral drug amantadine, probably have a

similar water-structuring effect, and amantadine, which is widely used as a therapy in Parkinson's disease, has an anti-excitotoxic action.

Article continued in Part 2 - [click here](#)

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V. V. Zakusov and R. U. Ostrovskaya, "Increased resistance of mice to hypoxia under the influence of tranquilizers of the benzodiazepine series," Byulletan Eksperimentalnoy Biologii i Meditsiny 71(2), 45-47, 1971. [The protection was not from the sedative effects, "Rather, the protective effect of these compounds is attributed to some specific intervention in the metabolism whereby the sensitivity of the tissues to oxygen insufficiency is reduced. ...the cortical structures of the brain especially appear to derive enhanced resistance to oxygen deficiency."]

A. I. Zotin, "Aging and rejuvenation from the standpoint of the thermodynamics of irreversible processes," Priroda 9, 49-55, 1970. [The process of aging "...is manifested by a decrease in entropy and...also by a continuous decrease in the rate of heat production.. The organism exhibits two types of approaches to a steady state: (i) constitutive movement of the system to the final steady state and (ii) inducible return of the system to the current steady state after deviating under the influence of internal or external factors. Oogenesis represents a constitutive deviation from the steady state; entropy reaches a level sufficient for the start of development and passage of the living system into the state of constitutive approach to the final steady state. From the standpoint of the thermodynamic theory of development, oogenesis reflects the process of regeneration of the system. In all other stages of life there is only the aging process accompanied by a decrease in entropy."

M. M. Tikhomirova, et al., "Mechanisms underlying the resistance of genetic material of the animal cell to stress treatment," Genetika 30(8), 1092-1104, 1994. "...these studies prove that the formation of a mutation is a multistage process involving many cell and organism systems...which are affected by environmental factors.... They can hinder or accelerate the mutational process, in this way providing both a superadditive effect and adaptive response. Recent studies deal with a universal system of heat shock proteins, which is involved in the maintenance of resistance of genetic material and genetic processes in the cell." Gross, "Reproductive cycle biochemistry," Fertility & Sterility 12(3), 245-260, 1961. "The maintenance of an environment conducive to anaerobic metabolism--which may involve the maintenance of an adequate supply of the substances that permit anaerobiosis...seems to depend primarily upon the action of estrogen." "Glycolytic metabolism gradually increases throughout the proliferative phases of the cycle, reaching a maximum coincident with the ovulation phase, when estrogen is at a peak. Following

this, glycolysis decreases, the respiratory mechanisms being more active during the secretory phase. Eschbach and Negelein showed the metabolism of the infantile mouse uterus to be less anaerobic than that of the adult. If estrogen is administered, however, there is a 98 per cent increase in glycolytic mechanisms." "The effect of the progestational steroids may be such as to interfere with the biochemical pattern required for support of this anaerobic environment."

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K. Savolainen, et al., "Phosphoinositide second messengers in cholinergic excitotoxicity," *Neurotoxicology* 15(3), 493-502, 1994. "Acetylcholine is a powerful excitotoxic neurotransmitter in the brain. By stimulating calcium-mobilizing receptors, acetylcholine, through G-proteins, stimulates phospholipase C and cause the hydrolysis of a membrane phospholipid...." "Inositol-1,4,5-triphosphate is important in cholinergic neuronal stimulation, and injury. Cholinergic agonists cause tonic-clonic convulsions which may be either transient or persistent. Even short-term cholinergic convulsions may be associated with neuronal injury, especially in the basal forebrain and the hippocampus. Cholinergic-induced convulsions also elevate levels of brain calcium which precede neuronal injury. Female sex and senescence increase the sensitivity of rats to cholinergic excitotoxicity." "Furthermore, glutamate increases neuronal oxidative stress...."

L. N. Simanovskiy and Zh. A. Chotoyev, "The effect of hypoxia on glycogenolysis and glycolysis rates in the rat brain," *Zhurnal Evolyutsionnoy Biokhimii i Fiziologii* 6(5), 577-579, 1970. "Glycogenolysis and glycolysis in the whole brain of young and old rats were studied at sea level and under hypoxic conditions in a low-pressure chamber or at an altitude of 3,200 meters. The rate of carbohydrate metabolism increased during postnatal development. In the absence of hypoxia, the rate of accumulation of lactate from either glycogen or glucose increases with maturation of the animals. The brain of young rats consumes primarily glycogen, particularly under anaerobic conditions." "Adaptation of mature rats to intermittent hypoxia is related to an increase in glycolysis, whereas adaptation of rats to high altitudes results in an increase in glycogenolysis. The type of carbohydrate metabolism is thus similar to the metabolism characteristic of the early stages of ontogenesis."

Ye. Sadovskiy, "For the prolongation of human life," *Sovetskaya Belorussiya* 23, page 4, Dec. 1970. "...the accumulation of metals in the organism with age is one of the most important factors in the development of the aging process."

Cerebral ischemia (and several other imbalances, relating to steroid regulation, shock) might be relieved by naloxone: D. S. Baskin and Y. Hosobuchi, *Lancet* ii, 272-275, 1981.

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